

One building. One city. Any city.



AI-based green remodeling optimization platform

— diagnosis, improvement scenarios and cost estimation for aged buildings, fully automated.

Measured 33.79% energy saving · verified to ASHRAE Guideline 14 · 178 real projects

GROWTH Co., Ltd.

Research Institute Company, Dong-Eui Univ.

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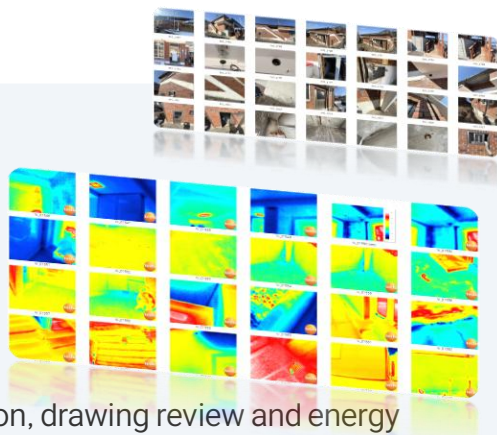


Every city faces the same wall.

From diagnosis to cost estimation, everything is still manual — a problem shared across borders



Manual survey



Site inspection, drawing review and energy simulation are still performed one expert at a time.



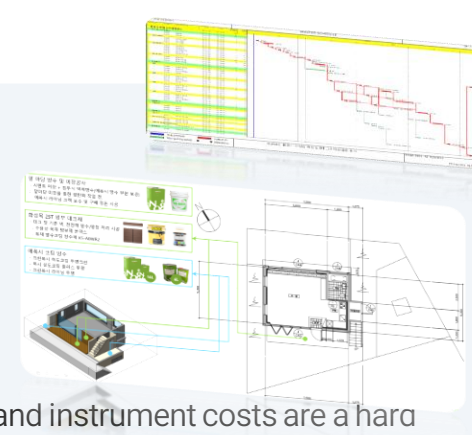
Judgment by experience



Improvement combinations rest on the engineer's judgment. An optimum is never guaranteed.



Cost of expertise



Analysis fees and instrument costs are a hard entry barrier for small building owners.

24.7%

Buildings' share of GHG

Korea, national inventory

2 weeks +

Manual lead time

Diagnosis, scenarios and cost for one building

< 1%

Owners who voluntary act (Green Remodeling)

Private owners starting green remodeling voluntarily

Ten years of R&D became a company.

A decade of national R&D is now an automated platform in service

Company profile

Company **GROWTH Co., Ltd.**
Green Remodeling Optimization & Wellness Technology Hub

CEO **Prof. Kyung-Soon Park**
Dept. of Architectural Engineering, Dong-Eui University

License **Registered GR provider (MOLIT)**
Ministry of Land, Infrastructure and Transport

Founded **2023 · 3rd year**
Professor-founded university spin-off

Status **Research institute company**
Industry-Academic Cooperation Foundation, Dong-Eui University

Stage **GROS in service · SGGP prototype done**
GROS : TRL 9 , SGGP : TRL 6 (*Technology Readiness Level)

2027 SGGP city pilot
2025 GROS completed
2023 Spin-off company
2016 R&D started



11



IP rights

2 registered · 8 filed · 1 software copyright

16



Standard models

Clustered from 178 measured projects

24



Own instruments

We collect our own field data

5



Government R&D

As principal investigator : approx. KRW 1.08bn

We do not sell software. We run a **closed loop**.

Diagnosis, prescription, treatment and follow-up run as one connected loop



01

DIAGNOSE

24 on-site instruments — thermal, airtightness, IAQ, etc.

Aging diagnosis through on-site expert inspections



02

PRESCRIBE

AI optimal combination of cost, energy, carbon.

Deriving Improvement Proposals by Energy Characteristics



03

TREAT

Matching with verified design and build partners.

National-certified Green Remodeling Provider Matching



04

MONITOR

Post-improvement verification and reporting.

Monitoring for continuous energy savings

Four Building Medic solutions Exterior · Interior · Renewable · Equipment — delivered together or one at a time



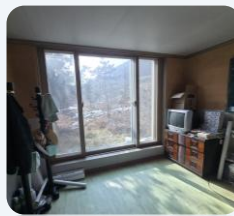
Exterior



The skin barrier — a fundamental cure for the aging envelope.



Interior



Renewing the aging interiors — comfortable and healthy green design.



Renewable



Empowering the building to generate its own energy.



Equipment



Improving aging - equipment systems to top performance

* GBMP (Green Building Medic Platform) — total service from precision diagnosis to design, construction and supervision, not brokerage.

Five stages, **one contract**.

From online intake to supervision and performance verification — under a single contract, continuous building performance optimization



1

Online intake

Automatic pre-diagnosis via GROS — aging risk and expected cost range.



2

On-site precision

24 specialist instruments produce a quantitative condition report.



3

Optimization

Cost–benefit simulation of improvement options with payback period.



4

Partner matching

Verified partner pool, or direct-management construction if preferred.



5

Supervision

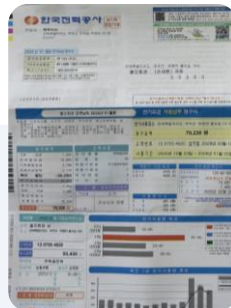
Quality inspection by process and post-completion performance report.



PAIN · Energy waste

High heating and cooling bills

→ Optimal improvement package



PAIN · Building defects

Condensation, mould, cracks and leaks

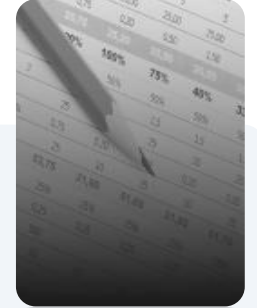
→ Verified performance figures



PAIN · Distrust in quotes

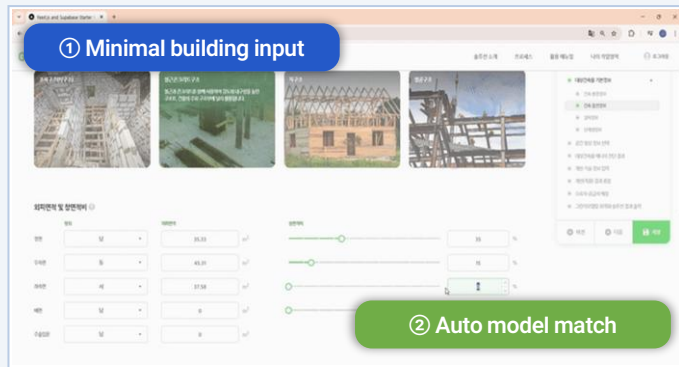
No basis on which to judge a quotation

→ Matching and supervision service

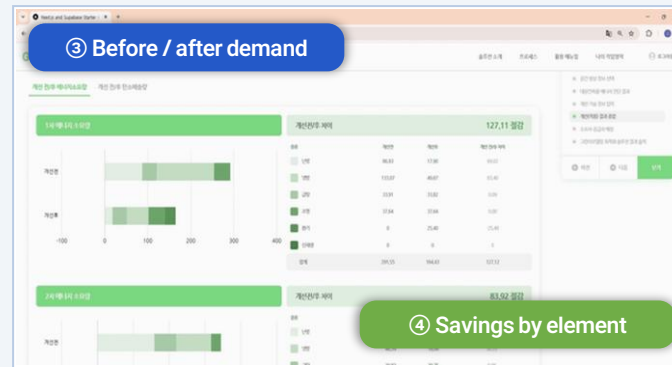


One input — diagnosis, scenarios and cost in **five minutes**.

One entry of building data returns diagnosis, scenarios, construction cost and asset value



Input — type, envelope, standard model



Result — before / after and savings

GROS · Key specification

Target

Detached houses · small commercial

User

Owners · SME designers & builders

Engine

AI genetic algorithm optimizer

Status

Completed · in service

33.79%

Energy saving

Measured · ASHRAE Guideline 14

68% ↓

Analysis cost

vs. conventional expert workflow

90% ↓

Lead time

2 weeks+ → 5 minutes

* GROS (Green Remodeling Optimization Solution) — AI genetic algorithm derives the optimal combination of cost, energy and carbon.



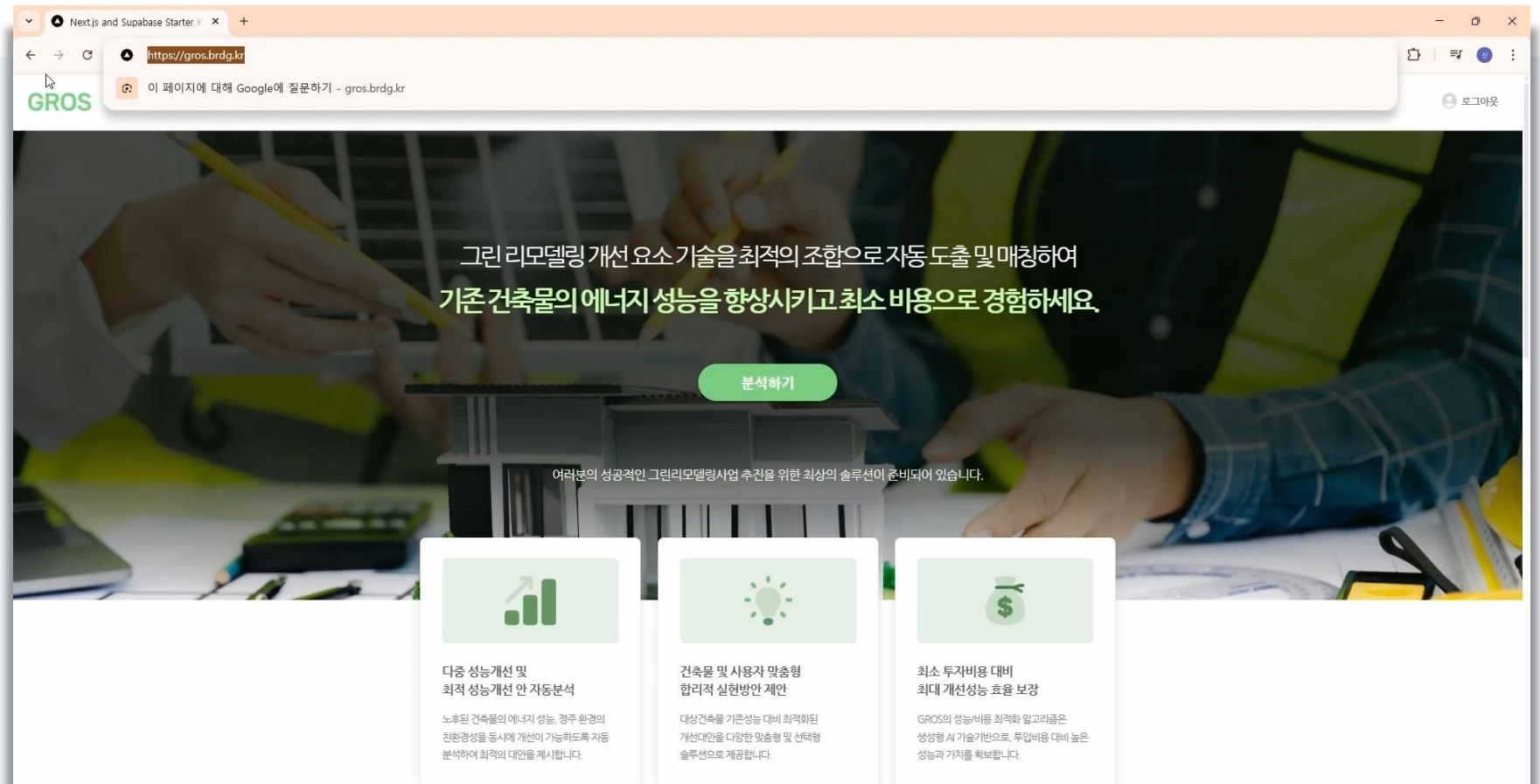
One input — diagnosis, scenarios and cost in five minutes.



G.R.O.S. solution demonstration video

The live web application — not a mock-up.

One entry of building data returns diagnosis, scenarios, construction cost and asset value



Execution of the GROS solution web page

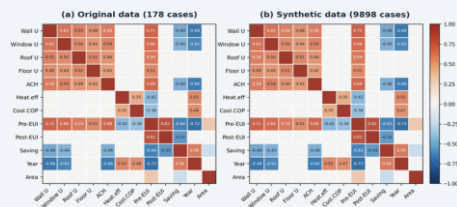
178 real projects, compressed into an engine.

178 measured projects became 16 standard models and an optimization algorithm



01

FIELD DATA

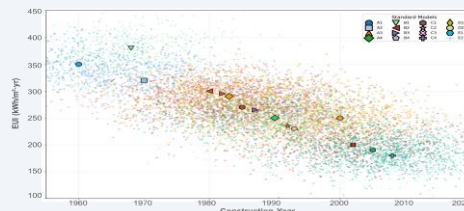


Measured performance from 178 real green remodeling projects.



02

STANDARD MODELS

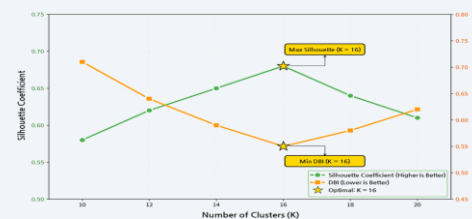


GA clustering by use, scale and building age — 16 models.



03

OPTIMIZER

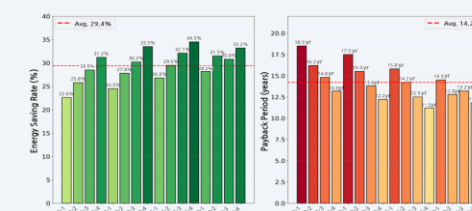


GA + PSO hybrid solving three objectives at once.



04

PARETO SET



Cost / energy / carbon trade-off and automatic reporting.

10,000

Monte-Carlo runs

Algorithm with a 95% confidence interval



4-layer

Human-in-the-loop review

AI output reviewed by engineers in four layers



SCI

SCI-level journal reviewed algorithm

Uncertainty quantification and standard-model matching



Measured, not modelled.

Performance proven by measurement, not by modelling alone

① MEASURED

✓ **33.79%**



Energy saving

- Measured before / after, Tongyeong H
- Verified to ASHRAE Guideline 14
- Payback 13–20 years

② PREDICTION

🎯 **±8.5%**



Mean prediction error

- Benchmarked against real GR projects
- Mid-size office buildings ±6.2%
- R^2 0.91 · RMSE 11.2 kWh/m²·yr
- 5-fold cross-validation

③ ALGORITHM

⚙️ **0.87**



Hypervolume

- Ahead of NSGA-II 0.81 and PSO 0.76
- Converges within 50 generations
- 45 s under 100 concurrent users

🎓 Published algorithm · Peer-reviewed in an SCI journal

“Uncertainty quantification and standard-model matching algorithm for energy performance improvement of small aged buildings”

178

Measured projects

16

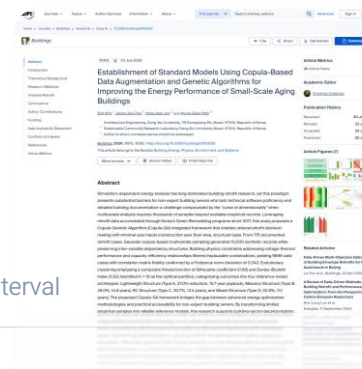
GA standard models

10,000

Monte-Carlo runs

95%

Confidence interval



⚠️ Under verification

Agreement between auto-estimated and actual construction cost is still being verified.

Construction cost agreement — verification in progress

From one building to an **entire city**.

Scaled to the district, matching demand and supply automatically

① City-scale input

연평균 온도 *
2007

연평균 (°C) *
1800

노후건축물 에너지 표준모델 **최적**
업무시설 · 중규모 표준모델 (2000년대)

소재지 (시도)
부산광역시

주용도
업무시설

구조형식
철근콘크리트 구조

최적화 모드
maximize_savings (에너지 절감 최대화)

개선 사례도 검토

표준모델 기반 최적 개선조합을 자동으로 도출합니다

최적의 대상 건축물 정보를 입력한 뒤, 노후건축물 에너지 표준모델과 특성합보 최적화 결과를 지정하고 추진 실행을 누르면 다중 알고리즘이 비용-에너지 성능을 동시에 분석합니다.

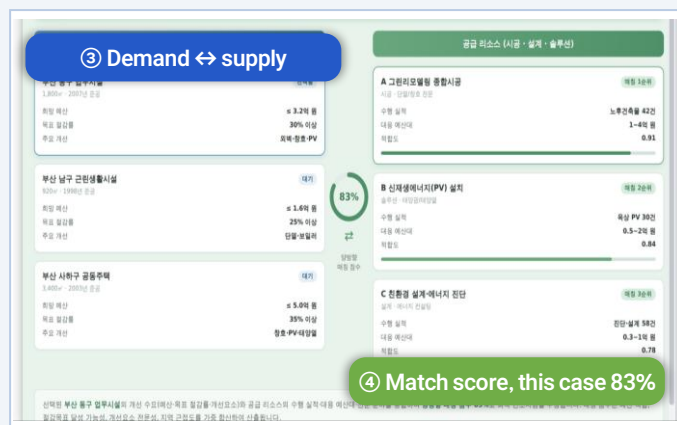
standard_model.csv 경로
C:\Users\GROSDesktop\standard_model.csv

특성합보 최적화결과 파일 경로
C:\Users\GROSDesktop\GR

표준모델 운영결과 파일 경로
C:\Users\GROSDesktop\GR

② Standard-model based

Input — city-scale data



Matching — demand ↔ supply

What SGGP does

- Rapid diagnosis of a whole city
- Three objectives at once
- Automatic demand-supply matching
- Reports generated instantly

86.1%

Match accuracy

Owners ↔ designers, builders, RE firms

3

Objectives at once

Cost · energy · carbon, optimized together

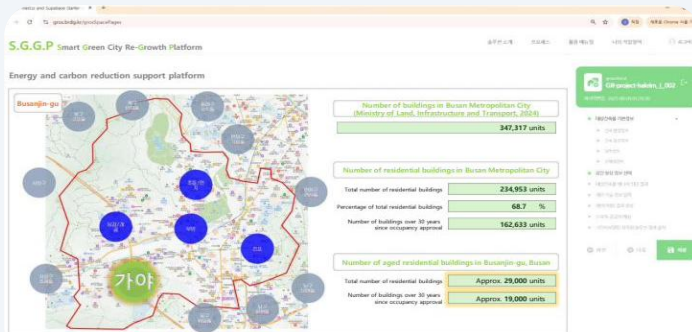
2027

City pilot ready

Prototype complete · Busan pilot in preparation

Busanjin-gu — 1,880 buildings, one simulation.

1,880 aged residential buildings diagnosed in a single pass, with no site survey



District-level monitoring interface

Scope of analysis

- Gaya-dong, Busanjin-gu, Busan
- Aged residential stock only
- Clustered on 16 standard models
- Diagnosis without on-site survey

What a city gets

- Which buildings to improve first
- How much carbon each option cuts
- How much budget it takes
- Evidence base for policy

1,880

Number of buildings analyzed

Aged residential stock, Gaya-dong

2,256 t

Expected CO₂ cut per year

Annual reduction, simulated

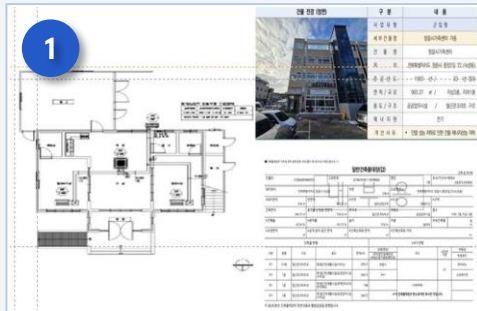
KRW 256.8 bn

Improvement target market

Estimated improvement construction volume

Not a demo — a consulting report in hand.

The deliverable is a report the owner can hold, not a screen



Survey & Drawing Review

건축물 용도 및 규모 정보(도면) 검토



On-site inspection

건축 구조 · 내외장재 · 설비 현장 측정조사



Economic Analysis

개선 대안 기술조합별 개략 공사비 산출



Performance & economics

에너지 성능 대비 경제성 최적화 분석

Swap the baseline data. That is the whole **localization**.

The engine stays unchanged; only the baseline data is replaced



01

MEASURE

On-site measurement with 24 own instruments.

자체 측정장비 사용 현장 실측 데이터 수집



02

EXPAND

178 verified cases expanded to 10,000 by Monte-Carlo.

실증 데이터 수집 → 머신러닝을 통한 데이터 확장



03

CLUSTER

GA clustering into standard models by use, scale and age.

용도·규모·연식별 군집분석을 통한 표준모델 도출



04

LOCALIZE

Replace climate, codes and unit prices — the engine is unchanged.

기후·법규·단가 데이터만 교체, 엔진은 그대로

✓ What stays the same

Optimization engine

Algorithm

Workflow

Report structure

🌐 What we replace

Climate data

Building codes

Construction unit prices

Standard models

The most aged city in Korea is our **live testbed**.

We prove it first where steep terrain, aging residents and aging stock overlap

BUSAN



60.7%

Highest ratio nationwide - 211,548 of 348,634 buildings
Average proportion of buildings over 30 years old

SEOUL



57.2%

Largest aged cluster in the capital area
Average proportion of buildings over 30 years old

NATIONAL AVG.



44.4%

National average as of the end of 2024
Average proportion of buildings over 30 years old

Priority zone — five inner-city districts, 200,000 buildings

Steep terrain, aging residents and aging stock overlap
— redevelopment is not feasible here, so remodeling is the only option left.

Sanbok-doro hillside districts — redevelopment not feasible



Source

MOLIT — National Building Statistics,
2023 and end-2024

Thank you.

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Dong-Eui University, Busan, Korea

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